Ferentinos, Konstantinos P. "Deep learning models for plant disease detection and diagnosis." *Computers and electronics in agriculture* 145 (2018): 311-318.

Title

Plant Diseases Detection using ml

Summary

A Convolutional Neural Network (CNN)–based model is used to detect plant diseases using leaf images. The dataset contains more than 8,000 images, which include both healthy and diseased leaves. Images were collected from 25 different plant species and cover more than 55 types of plant diseases. Due to the diversity of plants and disease patterns, detecting plant diseases through leaf images is a challenging task. However, the proposed model achieves high performance, with an accuracy of 99.53%.

Automatic plant disease detection plays an important role in modern agriculture, as it helps farmers identify diseases at an early stage and protect their crops from major losses. This system can be integrated into a mobile application, which is especially useful for farmers in remote areas. The application can provide real-time disease identification along with suitable treatment suggestions.

Machine learning and deep learning techniques are widely used for image processing and disease detection tasks. Among them, CNN models are considered the most effective for image-based analysis. In this approach, the CNN model is trained using supervised learning, where images are labeled as healthy or diseased. The model learns important features from leaf images and uses them to classify new images accurately.

During the training phase, the CNN model is trained on a large dataset, and its performance is evaluated during testing using unseen images. The experimental results show that the model performs exceptionally well, achieving accuracy close to 100% in some test cases. Deep learning techniques are used to optimize parameters and improve the overall performance of the model.

In conclusion, this work demonstrates that CNN-based deep learning models are highly effective for plant disease detection using leaf images. The model achieves an accuracy of 99.53%, proving its reliability and efficiency. Early disease detection using such systems can significantly help in preventing crop damage and improving agricultural productivity.

E. Moupojou et al., "FieldPlant: A Dataset of Field Plant Images for Plant Disease Detection and Classification With Deep Learning," in IEEE Access, vol. 11, pp. 35398-35410, 2023, doi: 10.1109/ACCESS.2023.3263042.

Title

FieldPlant: A Dataset of Field Plant Images for Plant Disease Detection and Classification With Deep Learning

Summary

Feeding the growing global population is becoming a serious challenge, especially as the world population is expected to reach nearly 10 billion by 2050. A major reason for food loss today is plant disease, which damages crops and reduces yield every year. A large portion of agricultural production is wasted because diseases are not detected early

Most existing plant disease detection systems are trained using laboratory image datasets, such as PlantVillage.

These images usually contain a single leaf with a plain background and controlled lighting. Although models trained on such datasets show very high accuracy during testing, they perform poorly in real farm conditions. This happens because real field images are much more complex, often containing multiple leaves, branches, soil, and uneven lighting. As a result, models trained only on laboratory images fail when applied in real agricultural environments.

To overcome these problems, the authors proposed a new dataset called FieldPlant. This dataset contains 5,170 real field images collected directly from plantations, mainly focusing on cassava, corn, and tomato crops. Each image was carefully annotated under the supervision of plant pathologists to ensure correctness. Since many images contain multiple leaves, the annotation process resulted in a total of 8,629 labeled leaf samples across 27 disease categories. This makes FieldPlant more realistic and reliable for training plant disease detection models.

In conclusion, this research highlights the importance of using real field data instead of laboratory images for plant disease detection. The FieldPlant dataset provides a strong foundation for developing practical and reliable disease detection systems that can be used by farmers in real conditions. Future work should focus on improving model robustness by combining image segmentation, model ensembling, and better handling of background noise to achieve accurate disease detection in real agricultural environments.

Mishra U, Pandey A, G L, K T. Deep learning-based disease detection in potato and mango leaves: a comparative study of CNN, AlexNet, ResNet, and EfficientNet. Sci Rep. 2025 Dec 24;16(1):2788. doi: 10.1038/s41598-025-32607-5. PMID: 41436834; PMCID: PMC12823699.

Title: Plant Disease detection using ML.

Summary:

In recent years, deep learning especially CNN has become a powerful tool for plant disease detection. Earlier methods relied on traditional image processing techniques, where features like color, texture, and shape were manually extracted and classified using algorithms. Although these approaches achieved moderate accuracy, they were limited because handcrafted features were not robust enough to handle real-world variations like lighting changes, complex backgrounds, and different leaf orientations.

The introduction of deep learning marked a major shift in plant pathology research. CNN-based models automatically learn hierarchical features directly from image, reducing the need for manual feature engineering. Early studies demonstrated that CNNs could achieve performance comparable to human experts when trained on controlled datasets like PlantVillage.

Overtime, researchers began exploring deeper and more advanced architectures EfficientNet. EfficientNet further enhanced performance by using compound scaling and attention mechanisms, achieving high accuracy with better computational efficiency.

Recent studies also focused on improving generalization and real-world applicability. Techniques such as transfer learning, data augmentation, attention mechanisms, and segmentation were introduced to address challenges like small datasets, visually similar diseases, and field condition variability. However, many studies concentrated on a single crop and did not systematically compare multiple architectures on multi-crop.

Datasets.

Therefore, there remains a need for comprehensive benchmarking of different deep learning architectures on diverse plant disease datasets. The current study addresses this gap by comparing CNN,EfficientNet on potato and mango.

Leaf disease datasets to evaluate accuracy, stability and generalization performance.

Shoaib, Muhammad, et al. "An advanced deep learning models-based plant disease detection: A review of recent research." *Frontiers in Plant Science* 14 (2023): 1158933

Title: Plant Disease detection using ML.

Summary:

Plant disease cause very large economic losses in agriculture every year. Modern AI techniques Deep learning model like CNN, GAN and computer Vision to detect disease early and accurate , instead of manual technologies. Use different method for imaging like RGB and Hyperspectral imaging.

Agriculture play a big role in the world and provide jobs for one billion people across the world. Countries like India , China and USA have a large cultivated area. Plant disease are a big problem for farming, they losses 10-16% crop every year. Most disease cause like Fungi, bacteria.Farmer needs advance early stage detection and cost friendly.

This review follow a high quality, relevant studies and standard method. Read papers were collected important database such as Scopus and IEEE. They also applied inclusion and exclusion criteria to ensure the accuracy and reliability.

Technologies used like Thresholding , edge detection and SVM for image processing and filter out disease area of the plants.

The review also discusses advanced models such as Recurrent Neural Networks for temporal analysis, Generative Adversarial Networks for data augmentation, and emerging vision-language models that enable zero-shot learning. Object detection frameworks like YOLO and Faster R-CNN are highlighted for their role in agricultural robotics and real-time disease monitoring.

Despite significant progress, several challenges remain. These include the lack of large, well-annotated datasets, poor model generalization across different crops and environments, high hardware costs, and limited interpretability of complex deep learning models. These issues restrict large-scale adoption in real-world farming scenarios.

Overall, this review bridges the gap between academic research and practical agricultural applications. It demonstrates that Deep Learning and Computer Vision technologies have the potential to transform precision agriculture by enabling early disease detection, reducing crop losses, and minimizing dependence on costly expert consultations. With further improvements in dataset availability, model interpretability, and cost-effective deployment, these technologies can play a crucial role in sustainable and intelligent farming systems.

Plant Disease Detection Using Machine Learning Techniques: A Comprehensive Approach. *Int. J. Environ. Sci.* **2025**, *11* (12s), 1268-1480.

Title: Plant Disease detection using ML.

Summary

Plant disease detection has become an important research area because plant diseases directly affect crop production, food supply, and farmers' income. Traditionally, farmers indentify diseases by observing plant leaves with their eyes. However, this method requires expert knowledge , takes time, and may not always be accurate. Due to these limitations, many researchers have starts using computer vision, ml and dl techniques to automatically detect plant diseases from leaf images.

Severa studies have shown that dl models, especially CNN are very effective for plant disease detection. Researchers have successfully used CNN models to classify plant diseases across multiple plant species using image datasets of leaves. These studies reported very high accuracy, showing that CNN can automatically learn important features from images without manual feature selection. Similarly, other researchers proposed deep convolutional arch. For multi-class plant disease classification, further proving the capability of deep learning in this field.

Apart from dl some researchers have focused on integrating ml with iot technologies. For example, one study proposed an early detection system for grape diseases using sensors to monitor environmental conditions such as temperature, humidity, and leaf wetness. The system used ml models to predict disease occurrence and even sent alerts to farmers through SMS. This approach highlights the importance of combining environmental data with image based analysis for better disease prediction.

Overall, the literature shows a clear evolution from traditional image processing methods to advanced dl and hybrid systems. Early methods relied heavily on manual feature extraction, while modern approaches allow models to learn features automatically from large datasets. Despite significant progress, challenges such as dataset diversity, real time deployment, and robustness under different environmental conditions still exist. Therefore, ongoing research continue to explore optimized hybrid frameworks that combine segmentation, feature extraction, classification, and deployment techniques to achieve more accurate and practical plant disease detection systems.
